

Chrono Rift – Operating Systems Project Report

Course: Operating Systems – Spring 2026

Project: Chrono Rift

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1. Project Summary

Chrono Rift is a multi-process battle simulation demonstrating Operating Systems concepts: shared memory IPC, process-shared synchronization, threaded workers, stamina-based scheduling, signal-driven effects, deadlock handling over artifact resources, and real-time terminal rendering.

2. Required Folder Structure

Submission layout matches course specification:

- ``submission/Dockerfile`` — container build environment (unchanged baseline).
- ``submission/Makefile`` — builds ``bin/arbiter``, ``bin/hip``, ``bin/asp`` (LIBS set for ncurses).
- ``submission/requirements.txt`` — optional extra apt packages.
- ``submission/arbiter/`` — arbiter sources; ``arbiter.cpp`` is main entry.
- ``submission/hip/`` — HIP sources; ``hip.cpp`` is main entry.
- ``submission/asp/`` — ASP sources; ``asp.cpp`` is main entry.
- ``submission/shared.h``, ``submission/shared.cpp`` — shared ABI and weapon helpers.

3. Requirements Coverage

3.1 Process Architecture and IPC

- Three isolated processes: Arbiter, HIP, ASP.
- POSIX shared memory as the single gameplay state channel.
- Process attach, PID registration, and orderly shutdown.

3.2 Concurrency and Synchronization

- ``pthread_mutex`` in shared memory for critical sections.
- ``sem_t`` in shared memory for action-ready signaling (non-blocking drain in arbiter loop).
- Worker threads in HIP and ASP; arbiter remains authoritative.

3.3 Scheduling, Signals, Deadlock, Memory

- Stamina-based discrete-time scheduler with deterministic tie-break.
- Wall-clock stun and signal-based ultimate pause (ASP SIGSTOP / SIGCONT + SIGALRM).
- Artifact graph: acquire, wait edges, circular-wait detection, forced release.
- Contiguous inventory slots, storage, SWAP_IN.

3.4 Build and Environment

- Makefile produces all three executables.
- Dockerfile supports reproducible build and demo environment.

4. Turnaround Analysis

Turnaround here means the elapsed path from when an entity becomes ready to act until that turn is fully committed and the next scheduling cycle can begin. The design keeps that path bounded and observable.

4.1 Readiness Phase

- Implemented in: ``arbiter/scheduler.cpp`` — ``cr_arbiter_scheduler_tick``.
- Stamina advances in whole-second steps until at least one alive, non-stunned entity reaches max stamina.
- If multiple entities are ready, lowest entity id wins (deterministic fairness).
- Turn state records ``active_entity_id``, ``active_is_player``, and ``action_committed = 0``.

4.2 Action Submission Phase

- HIP: ``hip/player_threads.cpp`` — workers submit to ``player_actions[player_id]`` only when ``active_entity_id`` matches.
- ASP: ``asp/npc_threads.cpp`` + ``asp/npc_logic.cpp`` — active NPC slot submits to ``npc_actions[slot]``.
- Semaphore posts on READY actions (``sem_player_action_ready`` / ``sem_npc_action_ready``) signal activity without blocking the arbiter.

4.3 Consumption and Commit Phase

- Implemented in: ``arbiter/scheduler.cpp`` — ``cr_arbiter_consume_active_action``.
- Valid READY action matching ``turn_seq`` and actor is applied; invalid or missing actions wait (NPC timeout can auto-skip).
- On success, stamina is spent and ``turn_seq`` advances (``commit_action`` or equivalent path per action type).

4.4 End-of-Turn and Next Cycle

- After dispatch, ``check_end_conditions`` evaluates win/lose.
- Arbiter main loop (``arbiter/arbiter.cpp``) runs at ~1 Hz; next tick may grant the following ready entity.
- Net effect: each committed turn completes in one consume pass; worst-case NPC stall is bounded by ``CR_NPC_TURN_TIMEOUT_SECONDS``.

5. Game Logic with Implementation Mapping

5.1 Initialization and Seeded Randomness

- Behavior: roll-derived default seed (``CR_SEED`` optional); RNG state drives NPC count and stat rolls.
- Implemented in: ``arbiter/arbiter.cpp`` (``resolve_game_seed``), ``arbiter/scheduler.cpp`` (``cr_arbiter_init_demo_entities``, ``next_rand``).

5.2 Player Stats

- Behavior: player HP from ``roll_number / random(3..4)`` per player; shared base damage from roll digit.
- Implemented in: ``arbiter/scheduler.cpp`` inside ``cr_arbiter_init_demo_entities``.

5.3 Combat Actions

- STRIKE / EXHAUST / USE_WEAPON / HEAL / SKIP / SWAP_IN / QUIT.
- Implemented in: ``arbiter/scheduler.cpp`` — ``cr_arbiter_consume_active_action`` and helpers.

5.4 Target and Weapon Rules

- Combat targets must be enemies; ``USE_WEAPON`` requires weapon actually equipped.
- Implemented in: ``arbiter/scheduler.cpp`` (``is_valid_combat_target``, ``entity_has_equipped_weapon``); HIP parsing in ``hip/player_threads.cpp``.

5.5 NPC AI

- Behavior: mixed targeting (soft/tank/random), heal when low, swap better weapon from storage, exhaust high-stamina players, prefer weapon when stronger than strike.
- Implemented in: ``asp/npc_logic.cpp``.

5.6 Artifacts and Deadlock

- Behavior: exclusive artifact locks; circular wait broken by stripping one side's artifacts.
- Implemented in: ``arbiter/deadlock.cpp``; weapon mapping in ``arbiter/scheduler.cpp`` (``artifact_id_from_weapon``).

5.7 Dynamic Eclipse

- Behavior: Eclipse artifact can enter the world at mid-progress (kill threshold).
- Implemented in: ``arbiter/scheduler.cpp`` — ``check_end_conditions`` (manifestation log + ``exists_in_world``).

5.8 Signals

- Stun expiry and ultimate pause/resume.
- Implemented in: ``arbiter/signals.cpp``.

5.9 Win / Lose / Quit

- Win: NPC kill count threshold; Lose: no players alive; Quit: quit action or session end paths.
- Implemented in: ``arbiter/scheduler.cpp`` (``check_end_conditions``); lifecycle in ``arbiter/arbiter.cpp``.

5.10 Rendering

- Dedicated render thread and TUI dashboards with mutex-held snapshots.
- Implemented in: ``arbiter/render.cpp``, ``hip/hip_tui.cpp``, ``asp/asp_tui.cpp``.

6. File Responsibilities (Concise)

6.1 Root

- ``Makefile`` — compile all binaries.
- ``Dockerfile`` — standard environment.
- ``shared.h`` / ``shared.cpp`` — ABI, weapon table, helpers.

6.2 Arbiter

- ``arbiter.cpp`` — main, shm, loop, cleanup, outcome print after UI teardown.
- ``scheduler.cpp`` — rules engine.
- ``signals.cpp`` — stun / ultimate.
- ``deadlock.cpp`` — artifact deadlock.
- ``render.cpp`` — arbiter HUD.

6.3 HIP

- ``hip.cpp`` — process main.
- ``player_threads.cpp`` — commands → ``player_actions[]``.
- ``input.cpp`` — stdin reader.
- ``hip_tui.cpp`` — optional ncurses UI.

6.4 ASP

- ``asp.cpp`` — process main.
- ``npc_logic.cpp`` — decisions.
- ``npc_threads.cpp`` — per-slot workers.
- ``asp_tui.cpp`` — optional ncurses UI.

7. Commands Reference

7.1 Native

- ``cd submission && make``
- ``./bin/arbiter`` then ``./bin/hip`` then ``./bin/asp`` (arbiter first).

7.2 Docker

- ``docker build -t chrono-rift:local .`` from ``submission/``
- ``docker run --rm -it -v "$PWD":/app -w /app chrono-rift:local bash`` then ``make`` and run binaries.

8. Screenshot Evidence (Insert in Sequence)

8.1 Initial State

8.2 Docker

```
root@f212ad374e39:/app
File Actions Edit View Help
CHRONO RIFT - Arbiter HUD
Kills 0/10 | NPCs alive 6/6 | Players 2/2 | phase=RUN run=1 paus
Turn seq 1 | NOW: Player-1 id=0 (PLAYER) | committed=0

PLAYERS (weapons = CR_WEAPON.* ids, st = storage count)
Player-1 [#####] 300/300 HP st 100/100
wpn:
Player-2 [#####] 400/400 HP st 100/100
wpn:
ENEMIES
NPC-4 [#####] 67/67 HP st 54/150
NPC-5 [#####] 175/175 HP st 50/150
NPC-6 [#####] 144/144 HP st 56/150
NPC-7 [#####] 193/193 HP st 56/150
NPC-8 [#####] 52/52 HP st 42/150
NPC-9 [#####] 90/90 HP st 36/150

BATTLE LOG (arrows k/j PgUp/Dn Home/r=newest End=oldest)
#1 Initial concurrent NPC count: 6
#2 Init Player Player-1: hp=300 dmg=11 spd=50 maxSt=100
#3 Init Player Player-2: hp=400 dmg=11 spd=50 maxSt=100
#4 Init NPC NPC-4: hp=67 dmg=10 spd=27 maxSt=150
#5 Init NPC NPC-5: hp=175 dmg=10 spd=25 maxSt=150
#6 Init NPC NPC-6: hp=144 dmg=10 spd=28 maxSt=150
#7 Init NPC NPC-7: hp=193 dmg=10 spd=28 maxSt=150
#8 Init NPC NPC-8: hp=52 dmg=10 spd=21 maxSt=150
#9 Init NPC NPC-9: hp=90 dmg=10 spd=18 maxSt=150
#10 Initial RNG seed: 12011191
#11 Turn 1 ready: actor=Player-1(0), dt=2s, ready=2

root@f212ad374e39:/app
File Actions Edit View Help
HIP - Player view (log: arrows/PgUp/PgDn Home=newest End=oldest)
Kills 0/10 NPCs 6/6 alive Players 2/2 turn#1 game=1
Active: Player-1 id=0 (you if id matches)
PARTY
Player-1 [#####] 300/300 st100/100
w:
Player-2 [#####] 400/400 st100/100
w:
ENEMIES
NPC-4 [#####] 67/67 st 54/150
NPC-5 [#####] 175/175 st 50/150
NPC-6 [#####] 144/144 st 56/150
NPC-7 [#####] 193/193 st 56/150
NPC-8 [#####] 52/52 st 42/150
NPC-9 [#####] 90/90 st 36/150

LOG
#1 Initial concurrent NPC count: 6
#2 Init Player Player-1: hp=300 dmg=11 spd=50 maxSt=100
#3 Init Player Player-2: hp=400 dmg=11 spd=50 maxSt=100
#4 Init NPC NPC-4: hp=67 dmg=10 spd=27 maxSt=150
#5 Init NPC NPC-5: hp=175 dmg=10 spd=25 maxSt=150
#6 Init NPC NPC-6: hp=144 dmg=10 spd=28 maxSt=150
#7 Init NPC NPC-7: hp=193 dmg=10 spd=28 maxSt=150
#8 Init NPC NPC-8: hp=52 dmg=10 spd=21 maxSt=150
#9 Init NPC NPC-9: hp=90 dmg=10 spd=18 maxSt=150
#10 Initial RNG seed: 12011191
#11 Turn 1 ready: actor=Player-1(0), dt=2s, ready=2

>>> YOUR TURN: Player-1 (id=0) turn_seq=1 <<<
1 STRIKE 2 EXHAUST 3 USE WEAPON 4 HEAL 5 SKIP 6 SWAP_IN
See shared.h CR_WEAPON.* for weapon numbers.

Enter choice:

root@f212ad374e39:/app
File Actions Edit View Help
[1] NPC-5 id=5 hp=175/175 st= 50/150 | - t= 0 w= 0 seq=
[2] NPC-6 id=6 hp=144/144 st= 56/150 | - t= 0 w= 0 seq=
[3] NPC-7 id=7 hp=193/193 st= 56/150 | - t= 0 w= 0 seq=
[4] NPC-8 id=8 hp= 52/ 52 st= 42/150 | - t= 0 w= 0 seq=
[5] NPC-9 id=9 hp= 90/ 90 st= 36/150 | - t= 0 w= 0 seq=

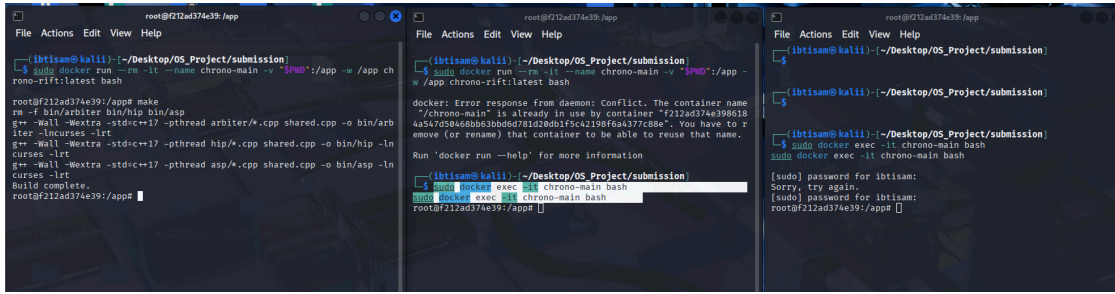
CR_ASP_TUI=0 disables this HUD[asp] PID registered. ncurses NPC H
UD on (CR_ASP_TUI=0 to disable).
[asp] NPC worker threads running (
one pthread per NPC slot).
```

A) Docker build

```
ibtisam@kali: ~/Desktop/OS_Project/submission
File Actions Edit View Help
└─$ sudo docker build -t chrono-rift .
[+] Building 2.2s (11/11) FINISHED                                docker:default
⇒ [internal] load build definition from Dockerfile                0.0s
⇒ ⇒ transferring dockerfile: 612B                                0.0s
⇒ [internal] load metadata for docker.io/library/ubuntu:22.04     2.0s
⇒ [auth] library/ubuntu:pull token for registry-1.docker.io       0.0s
⇒ [internal] load .dockerignore                                   0.0s
⇒ ⇒ transferring context: 2B                                       0.0s
⇒ [1/5] FROM docker.io/library/ubuntu:22.04@sha256:962f6cadeae0e 0.0s
⇒ [internal] load build context                                   0.0s
⇒ ⇒ transferring context: 38B                                       0.0s
⇒ CACHED [2/5] RUN apt-get update && apt-get install -y python3 0.0s
⇒ CACHED [3/5] COPY requirements.txt /tmp/requirements.txt        0.0s
⇒ CACHED [4/5] RUN grep -v '^#' /tmp/requirements.txt | grep -v 0.0s
⇒ CACHED [5/5] WORKDIR /app                                       0.0s
⇒ exporting to image                                              0.0s
⇒ ⇒ exporting layers                                              0.0s
⇒ ⇒ writing image sha256:4fb8fe8ac28bd4910bd1e44e34c5ca2466f0d1 0.0s
⇒ ⇒ naming to docker.io/library/chrono-rift                       0.0s

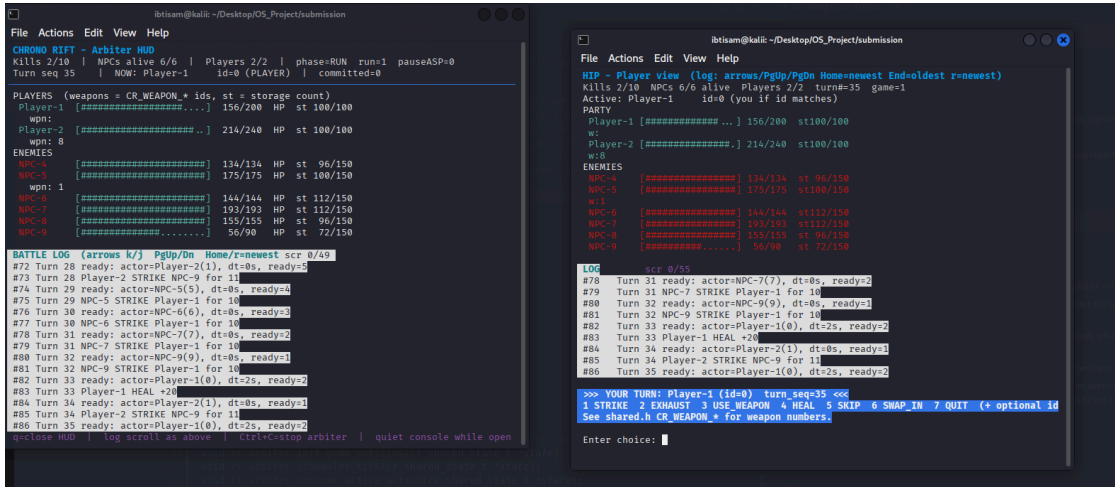
(ibtisam@kali)~[~/Desktop/OS_Project/submission]
└─$ sudo docker images
REPOSITORY              TAG         IMAGE ID      CREATED
SIZE
chrono-rift-env         latest      4fb8fe8ac28b  3 days ago
814MB
chrono-rift             latest      4fb8fe8ac28b  3 days ago
814MB
ibtisam1201/grade_calculator-app latest      5e4cb8cd33ba  11 days ago
1.54GB
hello-world             latest      e2ac70e7319a  6 weeks ago
10.1kB
```

B) Docker run / container shell



8.3 Gameplay (chronological)

A) Early logs



B) Mid logs


```
root@f212ad374c39:/app
File Actions Edit View Help
CHRONO RIFT - Arbitrer HUD
Kills 7/10 | NPCs alive 6/6 | Players 2/2 | phase=RUN run=1 pause
Turn seq 16 | NOW: Player-1 | id=0 (PLAYER) | committed=0

PLAYERS (weapons = CR_WEAPON_* ids, st = storage count)
Player-1 [.....] 270/300 HP st 100/100
Player-2 [.....] 400/400 HP st 100/100
w:8
w:8
ENEMIES
NPC-5 [.....] 44/67 HP st 135/150
NPC-6 [.....] 175/175 HP st 150/150
NPC-7 [.....] 144/144 HP st 150/150
NPC-8 [.....] 193/193 HP st 150/150
NPC-9 [.....] 155/155 HP st 96/150
NPC-9 [.....] 90/90 HP st 54/150

BATTLE LOG (arrows k/) PgUp/Dn Home=rnewest scr 0/10
#19 Turn 5 ready: actor=Player-1(0), dt=2s, ready=1
#20 NPC-NPC-8 defeated; dropped weapon=8
#21 Player-1 picked up weapon=8
#22 Turn 5 Player-1 STRIKE NPC-8 for 11
#23 NPC respawned: NPC-8 hp=155 dmg=14 wpd=10
#24 Turn 6 ready: actor=Player-2(1), dt=0s, ready=1
#25 Turn 6 Player-2 EXHAUST NPC-4 by 11 stamina
#26 Turn 7 ready: actor=NPC-5(5), dt=0s, ready=1
#27 Turn 7 NPC-5 STRIKE Player-1 for 10
#28 Turn 8 ready: actor=NPC-6(6), dt=0s, ready=1
#29 Turn 8 NPC-6 STRIKE Player-2 for 10
#30 Turn 9 ready: actor=NPC-7(7), dt=0s, ready=1
#31 Turn 9 NPC-7 STRIKE Player-2 for 10
#32 Turn 10 ready: actor=NPC-4(4), dt=1s, ready=1
#33 Turn 10 NPC-4 STRIKE Player-1 for 10
#34 Turn 11 ready: actor=Player-1(0), dt=1s, ready=1
#35 Turn 11 Player-1 STRIKE NPC-4 for 11
#36 Turn 12 ready: actor=Player-2(1), dt=0s, ready=1
#37 Turn 12 Player-2 EXHAUST NPC-9 by 11 stamina
#38 Turn 13 ready: actor=NPC-9(9), dt=1s, ready=1
#39 Turn 13 NPC-9 STRIKE Player-1 for 10
#40 Turn 14 ready: actor=Player-1(0), dt=1s, ready=1
#41 Turn 14 Player-1 USE_WEAPON(8) on NPC-4 for 1
#42 Turn 15 ready: actor=Player-2(1), dt=0s, ready=1
#43 Turn 15 Player-2 HEAL +14
#44 Turn 16 ready: actor=Player-1(0), dt=2s, ready=1
q:close HUD | log scroll as above | Ctrl+C:stop arbitrer | quiet co

root@f212ad374c39:/app
File Actions Edit View Help
HIP - Player view (log: arrows/PgUp/PgDn Home=rnewest End=oldest
Kills 1/10 NPCs 6/6 alive Players 2/2 turn=16 game=1
Active: Player-1 | id=0 (you if id matches)
PARTY
Player-1 [.....] 270/300 st100/100
w:8
Player-2 [.....] 400/400 st100/100
w:8
ENEMIES
NPC-5 [.....] 44/67 st135/150
NPC-6 [.....] 175/175 st150/150
NPC-7 [.....] 144/144 st150/150
NPC-8 [.....] 193/193 st150/150
NPC-9 [.....] 155/155 st 96/150
NPC-9 [.....] 90/90 st 54/150

LOG scr 0/23
#24 Turn 6 ready: actor=Player-2(1), dt=0s, ready=1
#25 Turn 6 Player-2 EXHAUST NPC-4 by 11 stamina
#26 Turn 7 ready: actor=NPC-5(5), dt=0s, ready=1
#27 Turn 7 NPC-5 STRIKE Player-1 for 10
#28 Turn 8 ready: actor=NPC-6(6), dt=0s, ready=1
#29 Turn 8 NPC-6 STRIKE Player-2 for 10
#30 Turn 9 ready: actor=NPC-7(7), dt=0s, ready=1
#31 Turn 9 NPC-7 STRIKE Player-2 for 10
#32 Turn 10 ready: actor=NPC-4(4), dt=1s, ready=1
#33 Turn 10 NPC-4 STRIKE Player-1 for 10
#34 Turn 11 ready: actor=Player-1(0), dt=1s, ready=1
#35 Turn 11 Player-1 STRIKE NPC-4 for 11
#36 Turn 12 ready: actor=Player-2(1), dt=0s, ready=1
#37 Turn 12 Player-2 EXHAUST NPC-9 by 11 stamina
#38 Turn 13 ready: actor=NPC-9(9), dt=1s, ready=1
#39 Turn 13 NPC-9 STRIKE Player-1 for 10
#40 Turn 14 ready: actor=Player-1(0), dt=1s, ready=1
#41 Turn 14 Player-1 USE_WEAPON(8) on NPC-4 for 1
#42 Turn 15 ready: actor=Player-2(1), dt=0s, ready=1
#43 Turn 15 Player-2 HEAL +14
#44 Turn 16 ready: actor=Player-1(0), dt=2s, ready=1

>>> YOUR TURN: Player-1 (id=0) turn_seq=16 <<<
1 STRIKE 2 EXHAUST 3 USE_WEAPON 4 HEAL 5 SKIP 6 SWAP_IN 7
See shared.h CR_WEAPON_* for weapon numbers.

Enter choice:

CR_ASP_TUI=0 disables this HUD(asp) PID registered, ncurses NPC H
UD on (CR_ASP_TUI=0 to disable).
[asp] NPC worker threads running (
one pthread per NPC slot).
```

C) Late logs

```
itbisam@kali: ~/Desktop/OS_Project/submission
File Actions Edit View Help
CHRONO RIFT - Arbitrer HUD
Kills 6/10 | NPCs alive 6/6 | Players 1/2 | phase=RUN run=1 pauseASP=0
Turn seq 139 | NOW: Player-2 | id=1 (PLAYER) | committed=0

PLAYERS (weapons = CR_WEAPON_* ids, st = storage count)
Player-1 [.....] 0/200 HP st 50/100 DEAD
Player-2 [##.....] 13/240 HP st 100/100
w:6,4
w:6,4
ENEMIES
NPC-4 [.....] 111/156 HP st 24/150
NPC-5 [.....] 175/175 HP st 75/150
w:1,7
w:1,7
NPC-6 [##.....] 20/140 HP st 84/150
NPC-7 [.....] 193/193 HP st 112/150
NPC-8 [.....] 155/155 HP st 32/150
NPC-9 [.....] 157/157 HP st 144/150

BATTLE LOG (arrows k/) PgUp/Dn Home=rnewest scr 0/50
#296 Turn 132 Player-2 USE_WEAPON(8) on NPC-6 for 12
#297 Turn 133 ready: actor=NPC-6(6), dt=0s, ready=2
#298 Turn 133 NPC-6 HEAL +14
#299 Turn 134 ready: actor=NPC-7(7), dt=0s, ready=1
#300 Turn 134 NPC-7 STRIKE Player-2 for 10
#301 Turn 135 ready: actor=NPC-5(5), dt=1s, ready=1
#302 Turn 135 NPC-5 USE_WEAPON(7) on Player-2 for 40
#303 Turn 136 ready: actor=Player-2(1), dt=1s, ready=3
#304 Turn 136 Player-2 HEAL +24
#305 Turn 137 ready: actor=NPC-4(4), dt=0s, ready=2
#306 Turn 137 NPC-4 STRIKE Player-2 for 10
#307 Turn 138 ready: actor=NPC-8(8), dt=0s, ready=1
#308 Turn 138 NPC-8 STRIKE Player-2 for 10
#309 Turn 139 ready: actor=Player-2(1), dt=2s, ready=1
q:close HUD | log scroll as above | Ctrl+C:stop arbitrer | quiet console while open

itbisam@kali: ~/Desktop/OS_Project/submission
File Actions Edit View Help
HIP - Player view (log: arrows/PgUp/PgDn Home=rnewest End=oldest rnewest)
Kills 6/10 NPCs 6/6 alive Players 1/2 turn=139 game=1
Active: Player-2 | id=1 (you if id matches)
PARTY
Player-1 [.....] 0/200 st 50/100 X
w:6,4
Player-2 [##.....] 13/240 st100/100
w:6,4
ENEMIES
NPC-4 [.....] 111/156 st 24/150
w:1,7
w:1,7
NPC-5 [.....] 175/175 st 75/150
NPC-6 [##.....] 20/140 st 84/150
NPC-7 [.....] 193/193 st112/150
NPC-8 [.....] 155/155 st 32/150
NPC-9 [.....] 157/157 st144/150

LOG scr 0/50
#302 Turn 135 NPC-5 USE_WEAPON(7) on Player-2 for 40
#303 Turn 136 ready: actor=Player-2(1), dt=1s, ready=2
#304 Turn 136 Player-2 HEAL +24
#305 Turn 137 ready: actor=NPC-4(4), dt=0s, ready=2
#306 Turn 137 NPC-4 STRIKE Player-2 for 10
#307 Turn 138 ready: actor=NPC-8(8), dt=0s, ready=1
#308 Turn 138 NPC-8 STRIKE Player-2 for 10
#309 Turn 139 ready: actor=Player-2(1), dt=2s, ready=1

>>> YOUR TURN: Player-2 (id=1) turn_seq=139 <<<
1 STRIKE 2 EXHAUST 3 USE_WEAPON 4 HEAL 5 SKIP 6 SWAP_IN 7 QUIT (+ optional id
See shared.h CR_WEAPON_* for weapon numbers.

Enter choice:
```

8.4 End States

A) WIN

```
root@f212ad374e39:/app
(ibtisam@kalii)-[~/Desktop/OS_Project/submission]
$ sudo docker run --rm -it --name chrono-main -v "$PWD":/app -w /app ch
rono-rift:latest bash

root@f212ad374e39:/app# make
rm -f bin/arbiter bin/hip bin/asp
g++ -Wall -Wextra -std=c++17 -pthread arbiter/*.cpp shared.cpp -o bin/arb
iter -lncurses -lrt
g++ -Wall -Wextra -std=c++17 -pthread hip/*.cpp shared.cpp -o bin/hip -ln
curses -lrt
g++ -Wall -Wextra -std=c++17 -pthread asp/*.cpp shared.cpp -o bin/asp -ln
curses -lrt
Build complete.
root@f212ad374e39:/app# ./bin/arbiter
[arbiter][init] Initial concurrent NPC count: 6
[arbiter][init] Init Player Player-1: hp=300 dmg=11 spd=50 maxSt=100
[arbiter][init] Init Player Player-2: hp=400 dmg=11 spd=50 maxSt=100
[arbiter][init] Init NPC NPC-4: hp=67 dmg=10 spd=27 maxSt=150
[arbiter][init] Init NPC NPC-5: hp=175 dmg=10 spd=25 maxSt=150
[arbiter][init] Init NPC NPC-6: hp=144 dmg=10 spd=28 maxSt=150
[arbiter][init] Init NPC NPC-7: hp=193 dmg=10 spd=28 maxSt=150
[arbiter][init] Init NPC NPC-8: hp=52 dmg=10 spd=21 maxSt=150
[arbiter][init] Init NPC NPC-9: hp=90 dmg=10 spd=18 maxSt=150
[arbiter] shared memory created and initialized.
[arbiter] Initial RNG seed: 12011191
[arbiter] scheduler tick is active. Press Ctrl+C to stop.
[arbiter][scheduler] Turn 1 ready: actor=Player-1(0), dt=2s, ready=2
[arbiter] Victory: players defeated required number of enemies.
[arbiter] shutdown complete.
root@f212ad374e39:/app#
```

```
root@f212ad374e39:/app
(ibtisam@kalii)-[~/Desktop/OS_Project/submission]
$ sudo docker run --rm -it --name cr-hip -v "$PWD":/app -w /app chr
lc "/bin/hip"
docker: permission denied while trying to connect to the Docker
: //var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock
ar/run/docker.sock: connect: permission denied

Run 'docker run --help' for more information

(ibtisam@kalii)-[~/Desktop/OS_Project/submission]
$ sudo docker run --rm -it --name cr-hip -v "$PWD":/app -w /app chr
lc "/bin/hip"
Unable to find image 'chronon-rift:local' locally
docker: Error response from daemon: pull access denied for chron
es not exist or may require 'docker login': denied: requested ac
is denied

Run 'docker run --help' for more information

(ibtisam@kalii)-[~/Desktop/OS_Project/submission]
$ sudo docker exec -it chrono-main bash
$ sudo docker exec -it chrono-main bash
root@71f8483e29ac:/app# cd /app && ./bin/hip
[hip] attached to shared memory.
[hip] Victory: players defeated required number of enemies.
[hip] shutdown complete.
root@f212ad374e39:/app#
```

B) LOSS

```
(ibtisam@kalii)-[~/Desktop/OS_Project/submission]
$ bin/hip
[hip] attached to shared memory.
[hip] Game over: all player characters defeated.
[hip] shutdown complete.
```

```
(ibtisam@kalii)-[~/Desktop/OS_Project/submission]
$ bin/arbiter
[arbiter][init] Initial concurrent NPC count: 6
[arbiter][init] Init Player Player-1: hp=200 dmg=11 spd=50 maxSt=100
[arbiter][init] Init Player Player-2: hp=240 dmg=11 spd=50 maxSt=100
[arbiter][init] Init NPC NPC-4: hp=67 dmg=10 spd=27 maxSt=150
[arbiter][init] Init NPC NPC-5: hp=175 dmg=10 spd=25 maxSt=150
[arbiter][init] Init NPC NPC-6: hp=144 dmg=10 spd=28 maxSt=150
[arbiter][init] Init NPC NPC-7: hp=193 dmg=10 spd=28 maxSt=150
[arbiter][init] Init NPC NPC-8: hp=52 dmg=10 spd=21 maxSt=150
[arbiter][init] Init NPC NPC-9: hp=90 dmg=10 spd=18 maxSt=150
[arbiter] shared memory created and initialized.
[arbiter] Initial RNG seed: 12011191
[arbiter] scheduler tick is active. Press Ctrl+C to stop.
[arbiter][scheduler] Turn 1 ready: actor=Player-1(0), dt=2s, ready=2
[arbiter] Game over: all player characters defeated.
[arbiter] shutdown complete.
```

C) QUIT

```
root@f212ad374e39:/app# ./bin/arbiter
[arbiter][init] Initial concurrent NPC count: 6
[arbiter][init] Init Player Player-1: hp=300 dmg=11 spd=50 maxSt=100
[arbiter][init] Init Player Player-2: hp=400 dmg=11 spd=50 maxSt=100
[arbiter][init] Init NPC NPC-4: hp=67 dmg=10 spd=27 maxSt=150
[arbiter][init] Init NPC NPC-5: hp=175 dmg=10 spd=25 maxSt=150
[arbiter][init] Init NPC NPC-6: hp=144 dmg=10 spd=28 maxSt=150
[arbiter][init] Init NPC NPC-7: hp=193 dmg=10 spd=28 maxSt=150
[arbiter][init] Init NPC NPC-8: hp=52 dmg=10 spd=21 maxSt=150
[arbiter][init] Init NPC NPC-9: hp=90 dmg=10 spd=18 maxSt=150
[arbiter] shared memory created and initialized.
[arbiter] Initial RNG seed: 12011191
[arbiter] scheduler tick is active. Press Ctrl+C to stop.
[arbiter][scheduler] Turn 1 ready: actor=Player-1(0), dt=2s, ready=2
[arbiter] Game session ended (quit or client disconnect).
[arbiter] shutdown complete.
root@f212ad374e39:/app#

(ibtisam@kali)-[~/Desktop/OS_Project/submission]
└─$ sudo docker exec -it chrono-main bash
root@71f8483e29ac:/app# cd /app && ./bin/hip
[hip] attached to shared memory.
[hip] Victory: players defeated required number of enemies.
[hip] shutdown complete.
root@f212ad374e39:/app# ./bin/hip
[hip] attached to shared memory.
[hip] shutdown complete.
root@f212ad374e39:/app# ./bin/hip
[hip] attached to shared memory.
[hip] Game session ended (quit or client disconnect).
[hip] shutdown complete.
root@f212ad374e39:/app#
```

9. Work Division

Complete before submission.

Member Name	Roll Number	Primary Responsibilities
Ibtisam Shahzad	22i-1201	Arbiter core, scheduler, signals, deadlock, integration
Muhammad Atyab	22i-1191	HIP/ASP, UI, command paths, Docker demo, report

10. Conclusion

Chrono Rift satisfies the course architecture: three processes, shared memory IPC, synchronized concurrency, threaded clients, stamina scheduling with clear turnaround, signal-driven mechanics, and resource deadlock handling, packaged with Makefile and Docker support for demonstration.